DONALD B. HAVERY

Applied AI Engineer, Forward-Deployed Focus · Agentic AI in Production · Evaluation · RAG · Tool Calling and MCP · Python and TypeScript

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SUMMARY

I build agentic AI solutions that run inside a business and connect to its data. For each step of a workflow I decide what the agent handles, what ordinary code handles, and what goes to a person, and I evaluate AI outputs before anything ships. My production customer-service agent runs on a live small-business site with Stripe checkout. It quotes prices only through tool calls against the price book, and it hands any case it cannot close to a person with the reason and the transcript. I work in Python and TypeScript with LLMs, RAG pipelines, tool calling and MCP, and I build with Claude Code every day.

PRODUCTION SYSTEMS

Customer-service agent in production on a revenue-taking site

- Answers customer questions from the business's knowledge base and quotes from the price book by tool call. The model never states a price without a tool call.
- Every reply and every tool result passes the same guards before the customer sees it. A reply that contains a price with no tool call behind it is stopped and handed to a person.
- Human in the loop by design: the agent hands off on customer request, low confidence, repeated failure, frustration, prompt-injection flags, guard violation, undecided business policy or model error, and each handoff carries the transcript tail and the customer's contact when given.
- When a customer asks about a policy the business has not decided, the retrieval tool returns no answer and the case goes to a person. The rule is enforced in code, so the prompt cannot override it.
- Next.js, TypeScript, Claude API, Stripe, Vercel. Built and run solo.

Release gate for LLM and RAG output

Prompt assertions, retrieval precision and recall, citation grounding, comparison against a baseline. Exits non-zero on a regression, so a worse release fails CI. A mutation suite plants defects and confirms each one is caught. Python, MIT.

Agent loop with failure handling

Typed tool schemas, structured traces, bounded retry with backoff, a watchdog on every call, cost and step ceilings, an unsafe-action gate, and a six-class failure taxonomy. A recorded run shows a real hang cut off by the watchdog and a halt at the cost ceiling. Python, MIT.

LLM gateway with provider failover

OpenAI-compatible proxy with model routing, cross-provider failover behind circuit breakers, rate limiting, an exact-match response cache, token cost tracking, and an HNSW vector index written without a library. Python, MIT.

Policy-control evidence review

Maps control statements to supplied evidence and writes a verdict per control with the matched excerpt. Redacts secrets, keeps an append-only audit log, and leaves the decision with the reviewer. Python.

Also shipped

Compliance gap analysis app on Hugging Face Spaces that checks controls against SOC 2, GDPR and HIPAA and lists the gaps · weather workstation on a 3D globe for desktop and Android · CDL training site · fuel-delivery operations platform with dispatch, customer, driver, franchise and HQ roles · self-correcting text-to-CAD loop

TECHNICAL SKILLS

Agentic AI	Tool calling, Model Context Protocol (MCP) servers, multi-step agent loops, human-in-the-loop handoff, output guards
LLMs and Prompts	Claude, Gemini, OpenAI-compatible APIs, Qwen and Ollama local serving; prompt engineering and failure analysis
Evaluation	Fixed test sets, regression gates in CI, retrieval precision and recall, citation grounding, mutation testing
RAG and Data	RAG pipelines, hybrid retrieval (BM25 + dense), vector search indexes, ChromaDB, Postgres, SQLite, Redis
APIs and Integration	REST APIs, FastAPI, Next.js API routes, Stripe checkout, data modeling
Languages and Delivery	Python, TypeScript / JavaScript, Rust (Tauri); Docker, Vercel, GitHub Actions; Claude Code (CLAUDE.md, hooks, subagents, skills)

EXPERIENCE

Applied AI Engineer, Independent Current

- Build, deploy and maintain AI agents and applications for my businesses, from mapping the workflow to production support.
- Map each workflow before building: which steps an agent takes, which steps code takes, and where a person approves.
- Ship with automated tests and an evaluation gate, and keep a written record of each failure and its fix.

Line-Haul Driver, Old Dominion Freight Line Current

- Class A CDL with hazmat, tanker and triples endorsements. Company-certified driver trainer, certified through the carriers I have worked for. Daily work inside a freight operation run on dispatch and federal hours-of-service rules.

Founder and Operator

angrynirds, a web and AI studio for small business, live and taking payments, with the production agent above on its front desk · Trucking Tutor, a live CDL training site · a DTC grooming brand. Built and run alongside full-time work.

EDUCATION

Capella University, Master of Business Administration (in progress) | **Capella University**, B.S. Computer Science

CERTIFICATIONS AND RESEARCH

Anthropic teaching certificate, Teaching the AI Fluency Framework, plus **21 further Anthropic Academy certifications** across Claude, Claude Code, MCP, and enterprise deployment on Bedrock and Vertex AI

IBM, AI Product Manager Professional Certificate | **DeepLearning.AI**, Generative AI with LLMs

"Reference-Relative Believability", independent research paper on evaluating AI-generated media, archived on Zenodo. DOI 10.5281/zenodo.21138188